Mathematical Modeling

Simplex practice: a solved demonstration

Personal study notes

1 Task: two products and two resources

A workshop makes continuous quantities of products 1 and 2. One batch of product 1 uses two hours of cutting and one hour of finishing. One batch of product 2 uses one hour of cutting and two hours of finishing. Eight hours of each resource are available. Profits are 4 and 3 monetary units per batch. Determine the profit-maximizing quantities. Write standard form, choose an initial feasible basis, perform simplex by hand, and verify with MATLAB.

2 Define the model before calling the solver

Let $x_1, x_2 \geq 0$ be the numbers of batches. Fractional batches are allowed; there is no integer restriction. The LP is

$$\max z = 4x_1 + 3x_2, \qquad \begin{cases} 2x_1 + x_2 \leq 8 & \text{cutting,} \\ x_1 + 2x_2 \leq 8 & \text{finishing.} \end{cases}$$

Rows represent resources and columns represent products. Both inequalities are upper resource limits. Adding nonnegative unused-resource variables s_1, s_2 yields the equality/nonnegativity standard form

$$Sv = b, \qquad v \geq 0, \qquad \max c^T v,$$

where

$$v = \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \end{bmatrix}, \quad S = \begin{bmatrix} 2 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 8 \\ 8 \end{bmatrix}, \quad c = \begin{bmatrix} 4 \\ 3 \\ 0 \\ 0 \end{bmatrix}.$$

The highlighted columns form the identity. Choosing $B = (s_1, s_2)$ and setting $x_1 = x_2 = 0$ gives $v = (0, 0, 8, 8)^T$, a BFS. Phase I is unnecessary. This is not an optimum: it represents producing nothing with all resources unused.

3 Follow the hand simplex steps

Use the maximization convention $z_j = c_B^T \bar{a}_j$ and indicators $z_j - c_j$. A negative indicator identifies a potentially improving variable. Choose the most negative indicator; break ties by the smaller variable index. For positive entering-column coefficients, the ratios appear to the right of the tableau. The arrow marks the leaving row. Keep the variable headings fixed.

3.1 Initial tableau

c_B	Basic	x_1	x_2	s_1	s_2	RHS	Ratio ($x_1 \uparrow$)	Out
	c_j	4	3	0	0			
0	s_1	$\boxed{2}$	1	1	0	8	$8/2 = 4$	$\longrightarrow$
0	s_2	1	2	0	1	8	$8/1 = 8$	
	z_j	0	0	0	0	0		
	$z_j - c_j$	-4	-3	0	0	$z = 0$		

Enter x_1 and remove s_1. Apply $R_1 \leftarrow R_1/2$, then $R_2 \leftarrow R_2 - R_1$ using the new row 1. The basic coefficients become $c_B = (4, 0)^T$, so recompute the weighted row using four times row 1.

3.2 After the first pivot

c_B	Basic	x_1	x_2	s_1	s_2	RHS	Ratio ($x_2 \uparrow$)	Out
	c_j	4	3	0	0			
4	x_1	1	1/2	1/2	0	4	$4/(1/2) = 8$	
0	s_2	0	$\boxed{3/2}$	$-1/2$	1	4	$4/(3/2) = 8/3$	$\longrightarrow$
	z_j	4	2	2	0	16		
	$z_j - c_j$	0	-1	2	0	$z = 16$		

The point is $(x_1, x_2) = (4, 0)$. The remaining negative indicator means increasing x_2 improves profit while the basic variables adjust. Enter x_2 and remove s_2. Apply $R_2 \leftarrow (2/3)R_2$, then $R_1 \leftarrow R_1 - (1/2)R_2$ using the new row 2.

3.3 Final tableau and interpretation

c_B	Basic	x_1	x_2	s_1	s_2	RHS
	c_j	4	3	0	0	
4	x_1	1	0	2/3	$-1/3$	8/3
3	x_2	0	1	$-1/3$	2/3	8/3
	z_j	4	3	5/3	2/3	56/3
	$z_j - c_j$	0	0	5/3	2/3	$z = 56/3$

All indicators are nonnegative. Set the nonbasic variables $s_1 = s_2 = 0$:

$$x^* = (8/3, 8/3)^T, \qquad z^* = 56/3 \approx 18.666667.$$

Both resources are fully used. The objective equation is $z = 56/3 - (5/3)s_1 - (2/3)s_2$, so no feasible nonnegative slacks can improve it. Both coefficients are strictly positive; equality forces both slacks to zero, and the two independent resource equations then determine a unique optimum.

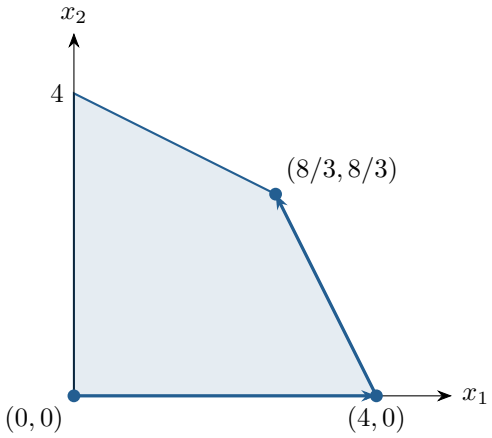


Figure 1: The hand path visits two edges. A solver may preprocess the model and use a different internal path while returning the same optimum.

4 Verify with Optimization Toolbox

Run `lab_03_demo.m` from the repository root. The script first reconstructs the three hand tableaux from their bases. This checks specified hand steps; it is not a general simplex implementation. It then calls `linprog` for both equivalent formulations. The solver minimizes, so negate the profit vector and negate its returned value:

```
options = optimoptions('linprog', ...
    'Algorithm', 'dual-simplex-highs', 'Display', 'none', ...
    'ConstraintTolerance', 1e-9, 'OptimalityTolerance', 1e-9);
A = [2, 1; 1, 2; -eye(2)];
b = [8; 8; 0; 0];
[x, fval, exitflag, output] = linprog([-4; -3], ...
    A, b, [], [], [], [], options);
% After checking exitflag, report profit as -fval.
```

The last two rows encode nonnegativity. The empty bound arguments are intentional; no sign restriction is left implicit. In the second call, use $-c$ as objective, $Sv = b$ as equalities, and $-I_4 v \leq 0$ as inequalities. Check the exit flag before interpreting the returned vector. Compare resource residuals, variable signs, profit and slacks with absolute tolerance 10^{-7}.

The selected solver uses dual simplex; the hand derivation above uses primal simplex. Preprocessing, basis selection and pivot rules can differ. A reported iteration count need not equal the two hand pivots, and an iteration log does not reproduce these tableaux. If a future assignment asks for an implementation of simplex itself, a call to `linprog` alone would not fulfill that task.

5 Sources and editorial remarks

This is an invented practice task, not an official assignment or a prediction of its exact format. The confirmed Lab 3 topics are simplex and standard form. The required solver, algorithm implementation and detailed assessment scope remain unconfirmed. Hand notation follows Chapter 3. All numbers and tableaux here were derived independently for this exercise and checked with exact rational arithmetic.

MATLAB syntax and the `dual-simplex-highs` option were checked against the official `linprog` reference and algorithm description on 25 October 2026. The companion requires Optimization Toolbox; the hand snapshot calculations use base MATLAB. Actual MATLAB execution and plot rendering remain unverified: the installed launcher exits with code 1 and no

output. Numerical values in this document are hand results, not claimed solver output. See `docs/matlab-verification.md` for the execution record.